

Paul (Chong Wa) Lai

Postdoctoral Researcher in Astrophysics
Institute for Theoretical Particle Physics and Cosmology, RWTH Aachen

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Research Interests

- Studying the cosmic-ray and gas distributions in the Milky Way with multi-messenger signals
- Probing astrophysical magnetic fields with polarisation and Faraday rotation
- Polarised radiative transfer in large-scale magnetised diffused media

Employment

2026 – present **Postdoctoral researcher** at RWTH Aachen University

Education

- 2022 – 2026 **Ph.D. in Astrophysics**
University College London, Mullard Space Science Laboratory,
Department of Space and Climate Physics
Thesis: Neutrinos and gamma-rays from the Milky Way Centre
Supervisors: Prof. Kinwah Wu, Dr. Matteo Agostini, and Prof. Foteini Oikonomou
- 2019 – 2022 **M.Phil. in Astrophysics**
The University of Hong Kong, Department of Physics
Thesis: Radio polarization study of the young pulsar wind nebula in supernova remnant G21.5–0.9
Supervisor: Dr. Stephen Chi-Yung Ng
- 2015 – 2019 **B.Sc. (1st class honours) in Theoretical Physics**
The Chinese University of Hong Kong, Department of Physics
Thesis: Behaviour of photons in modulating spacetime
Supervisors: Dr. Po Kin Leung and Prof. Kinwah Wu

Publications (*highlighted publications)

5. J. Y. H. Chan, A. Y. L. On, **P. C. W. Lai**, and Kinwah Wu, “Bridging Theory and Observation in the SKA Era: A Cosmological Polarized Radiative Transfer Framework for Point-to-Point Polarized Sky Comparisons”, Chapter in Advancing Astrophysics with the SKA – II (2026), [arXiv:2606.25105](#)
4. ***P. C. W. Lai**, Beatrice Crudele, Matteo Agostini, Hayden P. H. Ng, Ellis R. Owen et al., “Neutrinos as a New Tool to Characterise the Milky Way Centre”, [PRD Letter](#), **113**, L061306 (2026)
3. **P. C. W. Lai**, Chi-Yung Ng, Shumeng Zhang, “High-resolution Very Large Array Radio Observations of the Boomerang Pulsar Wind Nebula”, [ApJ](#), **1010**, 69 (2026)
2. ***P. C. W. Lai**, Kaye J. Li, Y. X. Jane Yap, Kinwah Wu and Albert K. H. Kong, “Spectropolarimetry of Synchrotron Radiation from Relativistic Electrons with Anisotropic Pitch-angle and Various Energy Distributions”, [MNRAS](#), Volume **542**, Issue **2**, Pages 902–916 (2025)
1. **P. C. W. Lai**, Chi-Yung Ng, N. Bucciantini, “Magnetic Field Structure and Faraday Rotation of the Plerionic Supernova Remnant G21.5–0.9” [ApJ](#), **930**, 1 (2022)

Conference Proceedings

2. **P. C. W. Lai**, B. Crudele, M. Agostini et al., “Neutrinos as a new tracer of the gas distribution in the Milky Way Centre”. [39th International Cosmic Ray Conference \(ICRC\)](#), id. 1089 (2025)
1. **P. C. W. Lai**, M. Agostini, F. Oikonomou et al., “Neutrino Imaging of the Galactic Centre and Millisecond Pulsar Population”, [38th International Cosmic Ray Conference \(ICRC\)](#), id. 1069 (2023)

Grants, Scholarships, Studentships and Awards

2025	RAS Grant – Conference Travel Subsistence (£1,000) , Royal Astronomical Society
2025	Research Student Conference Fund (£300) , Institute of Physics (IOP)
2025	MSSL Travel Grant (£500) , Mullard Space Science Laboratory, University College London
2024	Cosmoparticle Initiative Travel Grant (£2,000) , University College London
2023	MSSL Travel Grant (£200) , Mullard Space Science Laboratory, University College London
2022 – 2025	Graduate and Overseas Research Scholarship , University College London
2019 – 2021	Postgraduate Scholarship , Graduate School, The University of Hong Kong
2017 – 2019	The Dean's Honours List (3 times) , Faculty of Science, The Chinese University of Hong Kong
2019	C N Yang Prize , Department of Physics, The Chinese University of Hong Kong
2018	UCL MAPS Dean's Summer Student Scholarship , Faculty of Mathematics & Physical Science, University College London
2018	C N Yang Scholarship for Research Experience , Department of Physics, The Chinese University of Hong Kong
2017	Physics Prize , Department of Physics, The Chinese University of Hong Kong

Invited Talks

2026 Mar 4	SPIMAX Seminar Series , Oxford University, UK
2025 Aug 22	Seminar , Astronomical Institute, Tohoku University
2024 Jun 25	Seminar , Norwegian University of Science and Technology, Trondheim, Norway
2023 Dec 19–20	Invited Talk , the Energetic Universe and Multi-Messengers Workshop at National Taiwan Normal University
2023 Dec 15–17	Invited Talk , National Centre Theoretical Sciences (NCTS) Annual Theory Meeting: Particle Physics, String Theory, Cosmology, and Astrophysics at National Taiwan University
2023 Jul 25	Seminar , Department of Earth and Space Science, The University of Osaka

Conference and Workshop Presentations

2025 Jul 14-24	Contributed Talk , International Cosmic Ray Conference (ICRC 2025) in Geneva, Switzerland
2025 Jul 7-11	Contributed Talk , National Astronomical Meeting (NAM), Durham, UK
2024 Dec 9-13	Talk and Poster , Cosmic Rays and Neutrinos in the Multi-Messenger Era, Paris, France
2023 Oct 4–6	Contributed Talk , IV Gravi-Gamma-Nu Workshop in L'Aquila, Italy
2023 Jul 26–Aug 3	Poster Presentation , International Cosmic Ray Conference (ICRC 2023) in Nagoya, Japan

Teaching and Outreach Experience

2026	Sharing session speaker , Astronomical Training Programme for Secondary Students, Hong Kong
2025	Teaching Assistant of High Energy Astrophysics (SPCE0012) , Department of Space and Climate Physics, University College London
2024	Teaching Assistant of High Energy Astrophysics (SPCE0012) , Department of Space and Climate Physics, University College London
2024	Project Instructor (teaching high school students how to compose RGB images of astrophysical objects using real telescope data), MSSL Work Experience Week 2024, University College London
2021	Teaching Assistant of Advanced Astrophysics (PHYS4656) , Department of Physics, The University of Hong Kong
2021	Chief Laboratory Demonstrator , Department of Physics, The University of Hong Kong
2020	Teaching Assistant of Astronomy Laboratory (PHYS3660) , Department of Physics, The University of Hong Kong
2020	Teaching Assistant of Introductory Electricity and Magnetism (PHYS2255) , Department of Physics, The University of Hong Kong
2020	Certificate in Teaching and Learning in Higher Education , Centre for the Enhancement of Teaching and Learning, The University of Hong Kong

Awarded Telescope Proposals

- 2021 **Principal Investigator**, the Very Large Array (VLA)
Co-I: Ruolan Jin and Chi-Yung Ng
Project 21A-058: high resolution radio study of the Boomerang pulsar wind nebula (1.5 hr)
- 2021 **Co-Investigator**, the Very Large Array (VLA)
PI: Ruolan Jin, *Co-I*: Chi-Yung Ng and **Paul Chong Wa Lai**
Project 21A-265: probing the youngest pulsar wind nebula in the galaxy (1.5 hr)

Scientific Service

- Since 2026 **Referee** for American Astronomical Society (AAS) journals
- 2023 – 2024 **Seminar Organiser**, Department of Space and Climate Physics, University College London

Software/Code Experience

NIFTy (Bayesian inference), **GraphGP** (Gaussian process generator), **CPRT** (polarised radiative transfer), **DustEM** (interstellar dust emission), **AM3** (lepto-hadronic interactions), **CASA** (radio data reduction tool)

Languages

English (Proficient), Mandarin (Fluent), Cantonese (Native)